

Whole-Genome Sequencing Reveals Genotypic Resistance in Phenotypically Susceptible *Mycobacterium tuberculosis* Clinical Isolates

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Abstract

Background: Whole-genome sequencing (WGS) data of *Mycobacterium tuberculosis* (MTB) complex strains have revealed insights about genetic variants associated with drug resistance (DR). Rapid genome-based diagnostics are being sought for specific and sensitive identification of DR; however, correct prediction of resistance genotypes requires both informatics tools and understanding of available evidence. We analyzed WGS datasets from phenotypically susceptible MTB strains using MTB resistance identification software. **Methods:** WGS data for 1526 MTB isolates classified as phenotypically drug susceptible were downloaded from the ReSeqTB database. The TB-Profiler software was used to call Single Nucleotide Variants (SNV) associated with resistance to rifampicin (RIF), isoniazid (INH), ethambutol (EMB), pyrazinamide, fluoroquinolone (FLQ), streptomycin (STR), and aminoglycosides. The SNV were further matched against the 2021 World Health Organization (WHO) catalogue of resistance mutations. **Results:** Genome analysis of 1526 MTB strains susceptible to first-line drugs revealed 39 SNV associated with DR to be present in across 14 genes in 5.9% ($n = 90$) isolates. Further interpretation of SNV based on the WHO catalog of mutations revealed resistance that 21 (1.4%) MTB isolates were resistant to first-line (4 to RIF, 14 to INH, 3 to EMB) drugs. While, 36 (2.6%) isolates were resistant to second-line (19 to STR, 14 to FLQ, and three to capreomycin) agents. The most frequent predictive SNV were; *rpoB* Ser450 Leu for RIF; *katG* Ser315Thr, *inhA* Ser94Ala, *fabG1*-15C >T (for INH); *gyrA* Asp94Gly for FLQ; *embB* Met306 Leu for EMB; *rpsL* Lys43Arg for STR; and *tlyA* Asn236 Lys for Capreomycin. **Conclusions:** Our study highlights the value of WGS-based sequence data for identifying resistance in MTB. It also shows how MTB strains may be misclassified simply on phenotypic drug susceptibility testing, and that correct genome interpretation is key for correct interpretation of resistance genotypes that can be used to guide clinical treatment.

Keywords: Drug resistance, genotype, whole-genome sequencing

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INTRODUCTION

Tuberculosis (TB) remains a global public health concern, particularly in low-middle-income countries. According to the World Health Organization (WHO), an estimated 1.3 million deaths were reported in 2020.^[1] TB is treatable through a multi-drug regimen given for 4–6 months; however, treatment becomes more complex if there is any resistance to the first-line drugs; rifampicin (RIF), isoniazid (INH), pyrazinamide (PZA) and ethambutol (EMB). Hence, early identification of resistance in *Mycobacterium tuberculosis* (MTB) is important for the effective management of multi-drug resistant (MDR) TB, defined by resistance to RIF or INH. Classical methods of identification of drug resistance (DR) in MTB has been culture together with drug

susceptibility testing (DST) against different antibiotics, a process which may take weeks. If MDR or rifampin resistance by molecular methods is detected, both phenotypic and genotypic susceptibility testing for fluoroquinolones (FLQ) and aminoglycosides is indicated.^[2,3] DR identification has

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been advanced by the employment of molecular testing of genetic targets associated with DR. However, molecular methods of DR identification are challenging as more than one mutation with a particular gene or multiple genes may contribute to resistance. Therefore, sequencing methods such as whole-genome sequencing (WGS) or targeted next-generation sequencing are valuable for providing full genetic information on MTB isolates.

WGS of MTB can provide information on genes associated with resistance to first-line, second-line (FLQ, and aminoglycosides), as well as to new and repurposed drugs, including bedaquiline (BDQ), clofazimine, delamanid, and linezolid.^[4,5] Correct interpretation of WGS data is essential for making accurate predictions. The WHO TB Mutation Catalog 2021 provides guidance on single-nucleotide variations (SNV) with a high predictive value of resistance to anti-tuberculous drugs.^[6]

Robust methods for the interpretation of WGS data are required for the prediction of DR. SNVs associated with MTB DR can be interpreted by freely available bioinformatics tools such as TB-Profiler,^[5,7] PhyResSE,^[8] and KVARQ.^[9,10]

Although DST testing is the gold standard for the identification of DR in MTB, there may be discrepancies between phenotypes and genotypes. We investigated the likelihood of DR being missed in MTB isolates if they were simply categorized by routine DST. For this, we analyzed whole genomes of MTB strains classified as susceptible to first-line drugs RIF, INH, PZA, and EMB, and available in a public database. We used data from the ReSeqTB platform, which is a collaborative, curated knowledge base, designed to standardize and aggregate global MTB complex variant data from WGS with phenotypic DST and clinical data.

Genome data were analyzed using the standalone command line script of TB-Profiler, a pipeline that allows researchers to upload WGS sequence data to retrieve a report with information on lineage and predicts resistance across 14 anti-TB drugs.

METHODS

The study received approval from the Aga Khan University. It involved the analysis of publicly available genomes. No consent was required for this study.

We identified and downloaded full genome sequence data of MTB isolates. We first selected isolates with sequence read archive (SRA) ids and then filtered the isolates based on their susceptibility to first-line drugs (RIF, INH, PZA, and EMB) available from ReSeqTB https://platform.reseqtb.org/main/fullexport.html?navCode=PZ6_Rg.fnt01.g. SRA was used to obtain genome data through National Center for Biotechnology Information (NCBI) e-fetch utility from the E-Direct NCBI suite.

Resistome scanning was done using TB-Profiler as per the workflow in Supplementary Figure 1.

TBProfiler characterizes mutations from fastq files, that are mapped to MTB H37Rv reference genome (Genbank accession number: NC_000962.3).^[11] The SNV called by TB-Profiler algorithms was further analyzed in this study.

RESULTS

We identified 1526 MTB isolates identified as susceptible to first-line drugs (INH, RIF, EMB, PZA) in the ReSeqTB database. Genomic analysis of these isolates followed by SNV calling through TBProfiler analysis revealed the presence of SNV associated with a DR genotype in 90 (5.9%) samples. The SNV were found across 14 genes, comprising resistance to RIF (*rpoB*), INH (*katG*, *inhA*, *fabI*), PZA (*pncA*, *panD*), EMB (*embA*, *embC*, *embB*), streptomycin (STR) (*rpsL*, *rrs*, *gidB*), FLQ (*gyrA*) and capreomycin (*tlyA*) [Figure 1].

Description of single-nucleotide variations found in resistance-conferring genes

Genomic profiling of the MTB isolates identified 39 SNVs across 14 genes associated with resistance to seven different drugs [Supplementary Table 1]. As per this classification, 32 (6.3%) MTB genomes were found to carry SNVs in resistance-conferring genes associated with either to first line; INH, RIF, PZA, or EMB drugs. While, 67 (4.4%) isolates had SNV in genes associated with FLQ, CAP, and STR resistance.

As not all SNVs in target genes lead to DR, the challenge remains to make accurate predictions of genome data so that the resistance genotype can be determined. We used the database of mutations that was made available through a WHO mutation catalog initiative. It provides data on the predictive nature of mutations in MTB, using available evidence to develop criteria for sensitivity, specificity, and their positive predictive value.^[6,12]

SNVs were classified across to rules from the 2021 WHO Catalogue of MTB mutations.^[6] As shown in Table 1, reclassification of the SNV revealed that 57 (3.7%) isolates had high-confidence mutations. Resistance to first-line drugs was found in 21 isolates; four had *rpoB* mutations predictive of RIF resistance, 14 isolates with mutations in *katG*, *inhA*, and *fabG1* were predicted to be resistant to INH, and three isolates with SNV (*embB*) were resistant to EMB. Thirty-six isolates (2.6%) were resistant to second-line drugs; 14 had *gyrA* mutations associated with FLQ resistance, 3 isolates had *tlyA* mutations associated with resistance against capreomycin, and 19 had *rpsL* mutations associated with STR resistance. Of these, the most common resistance-predictive SNV found in each gene was; Ser450 Leu in *rpoB*, -15C <Tin *fabG1*, Ser315Thr in *katG*, Ser94A in *inhA*, Met306Ile in *embB*, Asp94Gly in *gyrA*, Asn236 Lys in *tlyA*, and Lys43Arg in *rpsL*.

DISCUSSION

WGS-based resistance genotyping is gaining attention as a tool for clinical TB. WGS offers an attractive option as it

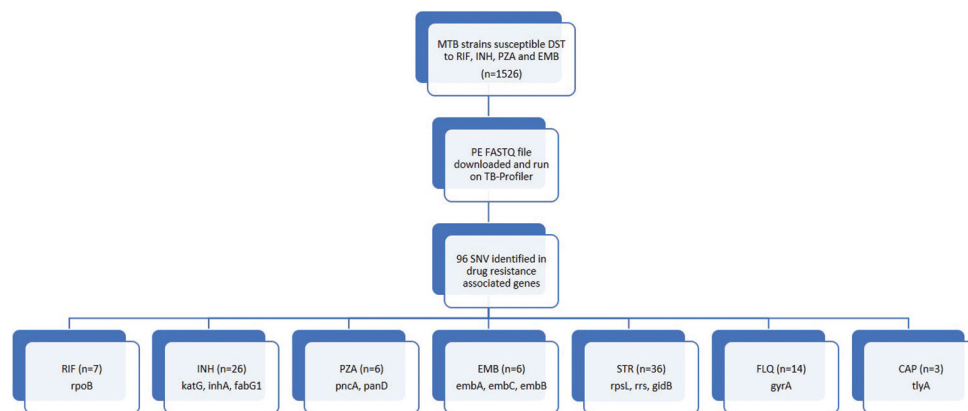


Figure 1: Resistance genotype identified in *Mycobacterium tuberculosis* strains with susceptible DST. Genome sequence data of 1526 MTB strains with a susceptible DST genotype were downloaded from the ReSeqTB database. Data were run on the TB-Profiler tool to identify a mutations in target genes. SNVs were identified in targets associated with genotypic resistance identified by mutations in key target genes for particular drugs are identified to RIF, INH, PZA, STR, EMB, FLQ and CAP. The number of isolates with mutations associated with a particular drug are noted together with genes screened in each case. RIF: Rifampicin, INH: Isoniazid, PZA: Pyrazinamide, STR: Streptomycin, EMB: Ethambutol, FLQ: Fluoroquinolone, CAP: Capreomycin, DST: Drug susceptibility testing, MTB: *Mycobacterium tuberculosis*, TB: Tuberculosis, SNV: Single nucleotide variations, PE FASTQ: Paired-End FASTQ

simultaneously examines all loci and provides information regarding both small and large changes in the genome. Advances in sequencing technologies and the accessibility of bioinformatics tools for the analysis of genome data have made sequence-based resistance identification more achievable for more laboratories. However, interpretation of resistance genotypes in MTB has been an evolving field and it is important to make correct interpretations to best guide treatment.

We conducted sequence analysis of MTB genomes with a susceptible DST to first-line drugs submitted in the ReSeqTB database. Genomes were investigated for the presence of DR associated SNV using the TB-Profiler software.^[5] We further interpreted the impact of the SNVs using the WHO mutation catalog.^[6]

Of the 1526 strains analyzed, 90 had mutations in at least one gene associated with resistance to anti-tuberculous drugs, of which 57 were SNV associated with high confidence mutations. Twenty-one isolates had resistance to first-line and 35 to second-line anti-tuberculous drugs. High-confidence mutations are confirmed by the WHO mutation catalog 2021.^[6] Therefore, there was a high predictive value of resistance-associated mutations in genes such as, *rpoB* for RIF, *katG*, *fabG1* and *inhA* for INH, *embB* and *embA* for EMB, *gyrA* for FLQ, *tlyA* for CAP, and *rpsL* for STR. Four MTB isolates were resistant to RIF, 14 to INH, and 3 to EMB. Further, resistance to FLQ was predicted in 14 isolates, that to capreomycin in three isolates and to STR in nineteen isolates.

The correct interpretation of MTB genome data is key for the prediction of resistance. Therefore, while available tools and pipelines for MTB genotype prediction are a convenient, easy-to-use platform for identifying SNVs in key genes associated, this information must be updated regularly. The MTB mutation catalog focuses on genotypes associated with DR in MTB, describing mutations, their specificity, sensitivity,

and validity in predicting phenotypic DR. However, MTB phenotypic resistance can remain unexplained even after investigating known genotypes associated with resistance.^[13]

The identification of a resistance genotype in strains with a susceptible DST may be surprising; however, neither phenotypic nor genotypic methods are perfect and there can be discordance between results, especially in the case of INH, FLQ and aminoglycosides.^[14-16]

Phenotype and genotype discrepancies may occur due to the limitations of DST. DST results can vary between different methods, and could be subject to technical variations but also the complexity of culturing MTB in antibiotics.^[17,18] Further, DST is usually performed only against four first-line drugs, and second-line drugs may be missed.

Importantly, in 36 isolates, a resistance genotype to second-line drugs (FLQ, CAP, and STR) was present, but the phenotypic profile was not known. This matches with previous reports that have shown MTB strains with an otherwise susceptible DST to have resistance to FLQ^[19] or BDQ.^[20] It raises the importance of sequenced-based identification for all drugs to ensure the most appropriate TB treatment.

CONCLUSIONS

Our study highlights the value of WGS-based resistance genotyping as increasingly important for the provision of resistance information. Further, it also informs on the value of sequencing to identify resistance genotypes in strains where phenotypic tests may miss DR. Therefore, resistance genotyping should be included in investigations of drug resistant MTB isolates especially, in high risk cases.

Ethical statement

This study was approved by the Ethical Review Committee of the Aga Khan University. Approval#3487-Pat-ERC-15.

Table 1: List of resistance conferring mutations found in genomes of MTB strains with a susceptible DST phenotype

Drug	Gene	No. of SNV	SNVs/mutation	No. of SNVs Occurrences	Isolate with SNV	Confidence grading [#]	Isolate with high confidence DR SNV	DR category
Rifampicin	<i>rpoB</i>	6	Ser450Leu	2	7	Assoc w R	4	First-line
			Leu430Pro	1		Uncert. Sig.		
			Ile491Phe	1		Assoc w R		
			Glu761Asp	1				
			Leu452Pro	1		Assoc w R		
			Val170Phe	1		Assoc w R		
Isoniazid	<i>fabG1</i>	3	c.-17G>T	2	17		14	First-line
			c.-8T>C	3		Uncert. Sig.		
			c.-15C>T	7		Assoc w R		
	<i>katG</i>	3	Thr394Ala	1		Uncert. Sig.		
			Ser315Thr	2		Assoc w R		
			Met257Ile	1		Uncert. Sig.		
Pyrazinamide	<i>inhA</i>	1	Ser94Ala	1	6	Assoc w R	0	First-line
	<i>panD</i>	1	Ile49Val	2				
	<i>pncA</i>	3	Asp136Asn	2		Uncert. Sig.		
			Ser18Pro	1		Uncert. Sig.		
Ethambutol	<i>embA</i>	5	c. 416_416del	1	6	Assoc w R	3	First-line
			c.-12C>T	1				
			Met306Ile	2		Assoc w R		
			Leu370Arg	1		Uncert. Sig.		
Fluoroquinolone	<i>embB</i>		Met306Leu	1	15	Assoc w R	14	Second-line
			Ala307Thr	1		Uncert. Sig.		
			Ala90Val	4		Assoc w R		
			Ala504Val	1				
	<i>gyrA</i>	6	Ser91Pro	2		Assoc w R		
			Gly88Cys	1		Assoc w R		
			Asp89Asn	2		Assoc w R		
			Asp94Gly	5		Assoc w R		
Capreomycin	<i>tylA</i>	1	Asn236Lys	3	3	Assoc w R	3	Second-line
Streptomycin	<i>rpsL</i>	4	Lys43Arg	15	36	Assoc w R	19	Second-line
			Lys88Arg	3		Assoc w R		
			Lys88Thr	2		Uncert. Sig.		
			Lys88Met	1		Assoc w R		
	<i>rrs</i>	2	r. 514a>c	8				
			r. 517c>t	2				
	<i>gid</i>	4	c. 102_102del	2				
			c. 4407793_4407880del	1				
			c. 115_115del	1				
			c. 559_572del	1				
Total number	14	39			90		57	

Data depicts SNVs identified in genomes of 1526 MTB isolates classified as susceptible in ReSeqDB based on phenotypic testing to named drug.

Confidence grading is from WHO Mutation Catalog 2021. SNV: Single nucleotide variant, DR: Drug resistance. FL: First line drug, SL: Second line drug.

Assoc w R: associated with resistance, Uncert. Sig: Uncertain significance

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Funding support was awarded to RH. ZH and RH designed the study. Bioinformatics analysis was done by SAR. AK conducted data analysis. Manuscript writing and editing were conducted by ZH, SR, AK, SS, and RH.

Conflicts of interest

There are no conflicts of interest.

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- technology for rapid detection of resistance to anti-tuberculous drugs. *Genome Med* 2019;11:41.
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Supplementary Table 1: List of mutations associated with a resistant genotype

Sample	FL	RIF	INH	PZA	EMB	STR	FLQ	CAP
ERR229992	FL	rpoB_p.Ser450Leu	-	-	-	-	-	-
ERR229987	SL	-	-	-	-	-	-	-
SRR2101293	SL	-	-	-	-	rpsL_p.Lys43Arg	-	-
SRR2101265	SL	-	-	-	-	-	-	-
SRR2100800	FL	-	-	-	embA_c.-12C>T	-	-	-
SRR2099962	FL	-	fabG1_c.-17G>T	-	-	-	-	-
SRR2100520	SL	-	-	-	-	rpsL_p.Lys43Arg	-	-
SRR2100641	SL	-	-	-	-	rpsL_p.Lys43Arg	-	-
SRR2100769	SL	-	-	-	-	rpsL_p.Lys43Arg	-	-
ERR400515	SL	-	-	-	-	-	gyrA_p.Asp94Gly	-
SRR2100684	SL	-	-	-	-	rpsL_p.Lys88Arg	-	-
ERR133973	FL	rpoB_p.Leu430Pro	-	-	-	-	-	-
SRR2101070	SL	-	-	panD_p.Ile49Val	-	-	-	-
SRR2100121	SL	-	-	-	-	-	gyrA_p.Asp94Gly, gyrA_p.Asp89Asn	-
SRR2100496	SL	-	-	-	-	-	-	-
SRR2100110	SL	-	-	-	-	rrs_r.514a>c	-	-
SRR2100675	SL	-	-	-	-	gid_c.102_102del	-	-
ERR144602	FL	rpoB_p.Ser450Leu	-	-	-	-	-	-
SRR2100469	FL	-	-	-	embB_p.Leu370Arg	-	gyrB_p.Ala504Val	-
SRR2100064	SL	-	-	-	-	rpsL_p.Lys88Met	-	-
ERR400505	SL	-	-	-	-	-	gyrA_p.Ser91Pro	-
ERR400534	SL	-	-	-	-	-	-	-
ERR551174	FL	-	fabG1_c.-17G>T	-	-	-	-	-
SRR2100522	SL	-	-	-	-	-	-	-
SRR2100807	SL	-	-	-	-	rpsL_p.Lys43Arg	-	-
SRR2100814	SL	-	-	-	-	-	gyrA_p.Asp94Gly	-
ERR144577	SL	-	-	-	-	gid_c.115_115del	-	-
ERR400413	FL	-	-	-	-	rrs_r.517c>t	-	-
SRR2100670	SL	-	-	-	embB_p.Met306Leu	-	-	-
SRR2100977	FL	-	-	-	-	rpsL_p.Lys43Arg	-	-
SRR2100619	SL	-	-	panD_p.Ile49Val	-	-	-	-
SRR2101164	SL	-	-	-	-	rpsL_p.Lys43Arg	-	-
SRR2100490	SL	-	-	-	-	rrs_r.514a>c	-	-
SRR2100611	FL	-	katG_p.Thr394Ala	-	-	rrs_r.514a>c	-	-
SRR2099928	FL	-	-	pncA_p.Asp136Asn	-	-	-	-
SRR2100959	FL	-	fabG1_c.-15C>T	-	-	-	-	-
SRR2100867	FL	rpoB_p.Ile491Phe	-	-	embB_p.Met306Ile	gid_c.102_102del	gyrA_p.Asp94Gly	-
SRR2100972	SL	-	-	-	-	rrs_r.514a>c	-	-
SRR2100710	SL	-	-	-	embB_p.Met306Ile	-	-	-

Contd...

Supplementary Table 1: Contd...

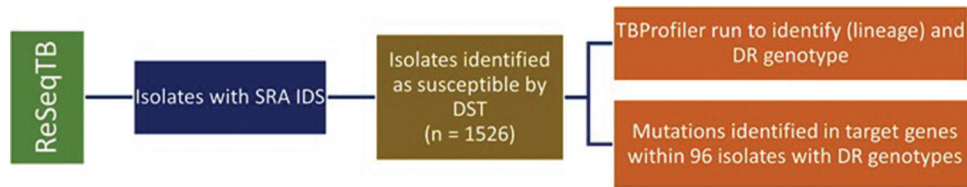
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ERR227998	SL	-	-	-	-	-	-	-
SRR2100086	SL	-	-	-	-	rpsL_p.Lys43Arg	-	-
SRR2100533	SL	-	-	-	-	gid_c.559_572del	-	-
SRR2099941	SL	-	-	-	-	-	-	-
SRR2101147	FL	-	-	pncA_p.Ser18Pro	-	-	-	-
SRR2101315	FL	-	katG_p.Ser315Thr	-	-	-	-	-
SRR2101135	SL	-	-	-	-	-	-	-
SRR2100824	FL	-	fabG1_c.-15C>T	-	-	-	-	-
ERR400491	SL	-	-	-	-	-	gyrA_p.Asp94Asn	-
SRR2100766	SL	-	-	-	-	rpsL_p.Lys88Thr	-	-
SRR2100924	SL	-	-	-	-	-	-	tlyA_p.Asn236Lys
SRR2100055	FL	-	-	pncA_p.Asp136Asn	-	-	-	-
SRR2100123	SL	-	-	-	-	rpsL_p.Lys43Arg	-	-
ERR551101	FL	-	fabG1_c.-15C>T	-	-	-	-	-
SRR2100527	SL	-	-	-	-	-	gyrA_p.Gly88Cys	-
SRR2100241	FL	-	-	pncA_c.416_416del	-	-	-	-
SRR2100307	FL	-	fabG1_c.-8T>C	-	-	-	-	-
SRR2101061	SL	-	-	-	-	-	-	-
SRR2100774	SL	-	-	-	-	-	gyrA_p.Ala90Val gyrA_p.Ala90Val, gyrA_p.Asp89Asn	-
SRR2100707	SL	-	-	-	-	-	-	-
SRR2100754	SL	-	-	-	-	rrs_r.514a>c	-	-
SRR2100926	SL	-	-	-	-	-	-	tlyA_p.Asn236Lys
SRR2100199	FL	-	fabG1_c.-15C>T	-	-	-	-	-
SRR2100668	SL	-	-	-	-	rpsL_p.Lys88Arg	-	-
SRR2100498	FL	-	katG_p.Ser315Thr	-	-	-	-	-
SRR2099990	FL	-	fabG1_c.-8T>C	-	-	-	-	-
SRR2101041	SL	-	-	-	-	-	-	tlyA_p.Asn236Lys
ERR038263	SL	-	-	-	-	rpsL_p.Lys43Arg	-	-
SRR2100541	SL	-	-	-	-	rpsL_p.Lys43Arg	-	-
SRR2100543	SL	-	-	-	-	rpsL_p.Lys88Arg	-	-
SRR2100279	SL	-	-	-	-	-	gyrA_p.Ala90Val	-
ERR228008	SL	-	-	-	-	-	-	-
SRR2101034	SL	-	-	-	-	rrs_r.514a>c	-	-
SRR2100708	SL	-	-	-	-	-	-	-
SRR2100511	SL	-	-	-	-	rpsL_p.Lys88Thr	-	-
SRR2099957	FL	-	inhA_p.Ser94Ala	-	-	-	-	-
ERR137278	FL	rpoB_p.Leu452Pro	-	-	-	-	-	-

Contd...

Supplementary Table 1: Contd...

Sample	FL	RIF	INH	PZA	EMB	STR	FLQ	CAP
SRR2100861	SL	-	-	-	-	-	gyrA_p.Ser91Pro	-
SRR2100756	SL	-	-	-	-	rpsL_p.Lys43Arg	-	-
SRR2100118	SL	-	-	-	-	rrs_r.514a>c	-	-
SRR2100170	FL	-	fabG1_c.-8T>C	-	-	-	-	-
SRR2100757	FL	rpoB_p.Val170Phe	-	-	-	-	-	-
SRR2100954	FL	-	fabG1_c.-15C>T	-	-	-	-	-
SRR2100314	SL	-	-	-	-	rrs_r.514a>c	-	-
SRR2100761	SL	-	-	-	-	-	gyrA_p.Asp94Gly	-
ERR137203	FL	rpoB_p.Glu761Asp	-	-	-	rrs_r.517c>t	-	-
SRR2100256	FL	-	katG_p.Met257Ile	-	-	-	-	-
SRR2100776	FL	-	fabG1_c.-15C>T	-	-	-	-	-
ERR228053	SL	-	-	-	-	-	-	-
ERR400537	SL	-	-	-	embC_p.Ala307Thr	-	-	-
SRR2101112	SL	-	-	-	-	-	-	-
SRR2100396	FL	-	fabG1_c.-15C>T	-	-	gid_Chromosome: g.4407793_440780del	-	-
SRR2100183	SL	-	-	-	-	-	gyrA_p.Ala90Val	-
SRR2100719	SL	-	-	-	-	-	-	-
SRR2100578	SL	-	-	-	-	rpsL_p.Lys43Arg	-	-
SRR2100559	SL	-	-	-	-	rpsL_p.Lys43Arg	-	-

FL: First line drug, SL: Second line drug, RIF: Rifampicin, INH: Isoniazid, PZA: Pyrazinamide, STR: Streptomycin, EMB: Ethambutol, FLQ: Fluoroquinolone, CAP: Capreomycin



Supplementary Figure 1: Description of study employing TB Profiler to identify drug resistance Phenotype Genotype discrepancies in Susceptible MTB isolates. DST: Drugss susceptibility testing, MTB: *Mycobacterium tuberculosis*, TB: Tuberculosis, SRA ID: Sequence read archiver identifier, DR: Drug resistance